



## 1 Background & Problem Statement

- The Silicon Sandbox:** FPGAs provide crucial hardware speed and software reprogrammability.
- Verification Bottleneck:** Testing complexity causes 70% of hardware projects to face delays and missed bugs. [1]
- Testing Dilemma:** Software simulations lack physical accuracy, while commercial hardware emulators are prohibitively expensive. [2]
- Proposed Solution:** A custom Python Hardware-in-the-Loop framework uses affordable development boards to directly test physical chips. [3]
- AI Optimisation:** Artificial Intelligence autonomously generates targeted test scenarios for a 16-bit ALU, achieving rigorous verification without expensive enterprise hardware.

| Method     | Accuracy | Speed    | Cost | Verdict      |
|------------|----------|----------|------|--------------|
| Simulation | Low      | Slow     | Low  | Unreliable   |
| Emulation  | High     | Fast     | High | Unaffordable |
| AI-HIL     | High     | Moderate | Low  | Viable       |

## 4 16-Bit Arithmetic Logic Unit

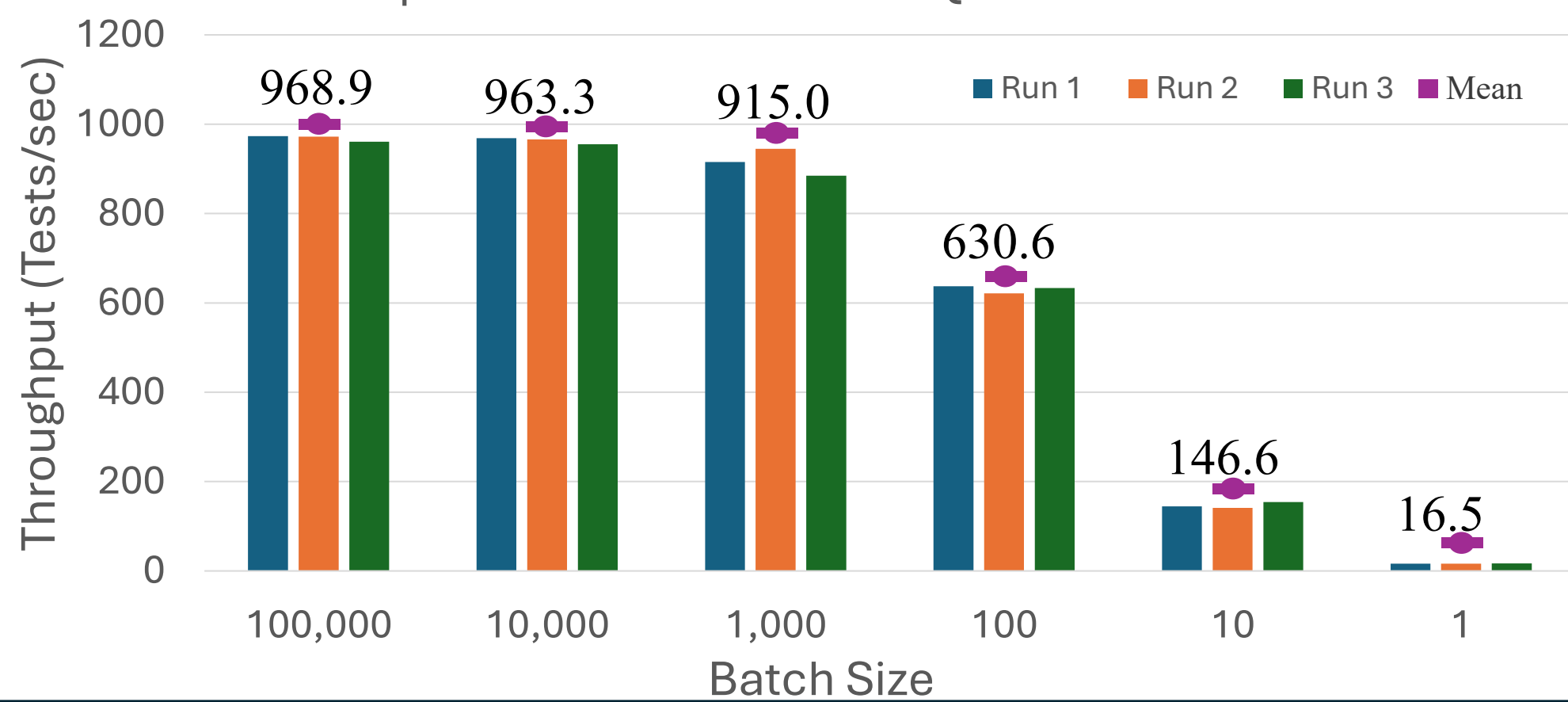
- Hardware:** Modular design with glitch-resistant 8N1 UART and 100 MHz to 1 kHz downsampled 7-segment display.
- Safeguards:** Divide-by-zero bypass and 4-slot internal memory.
- Execution:** 18 instructions. 1-cycle execution (delayed for complex arithmetic).

| Category        | Operations                         |
|-----------------|------------------------------------|
| Arithmetic      | ADD, SUB, MUL, DIV, MOD, SQRT, POW |
| Logic           | AND, OR, XOR, NOT                  |
| Shift/Mem/Unary | SHL, SHR, STO, RCL, INC, DEC, NEG  |

## 6 Results & Performance Data

- Peak Automation:** Achieved maximum mean throughput of 968.9 tests per second for large batch sizes (100k tests), 104.04 seconds @ 100% pass rate.
- AI Validation:** Beyond raw speed, the integrated AI successfully detected and diagnosed an intentionally injected Verilog logic error.
- Performance Scaling:** Clear bottleneck at smaller batch sizes. The FPGA executes instructions almost instantaneously; the drop to 16.5 tests per second for a single test is caused by the PC's Python file I/O and UART communication overheads

Tests per Second for Different Quantities of Tests



## 8 Risk Mitigation

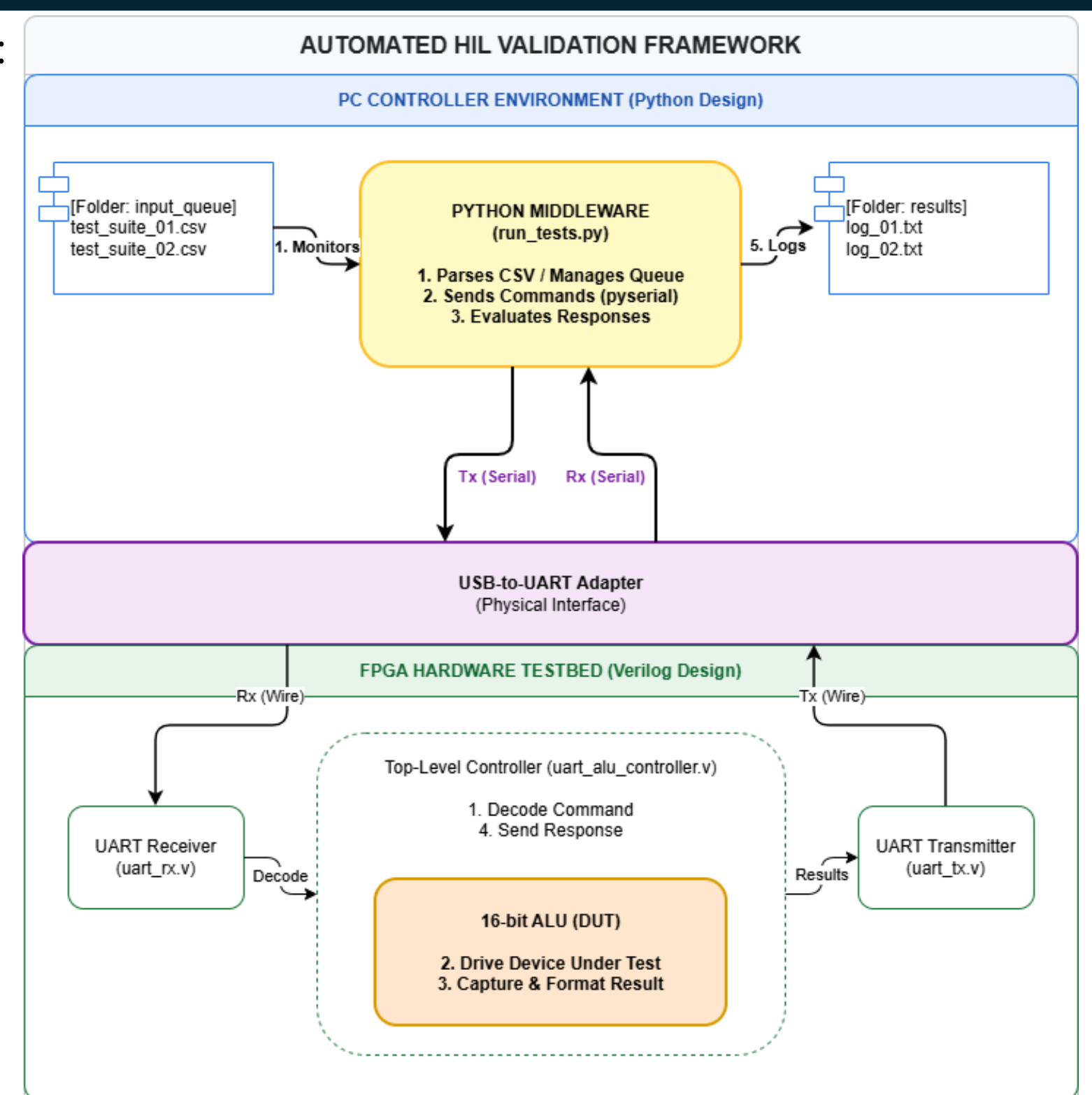
- Technical (High):** Integration bugs mitigated via modular Verilog simulation and incremental Python loopback testing.
- Safety (High):** DSE fatigue and physical trip hazards mitigated by mandated screen breaks and strict cable routing.
- Management (Medium):** Schedule creep and data loss mitigated through strict Gantt tracking and cloud version control.

## 2 Project Planning

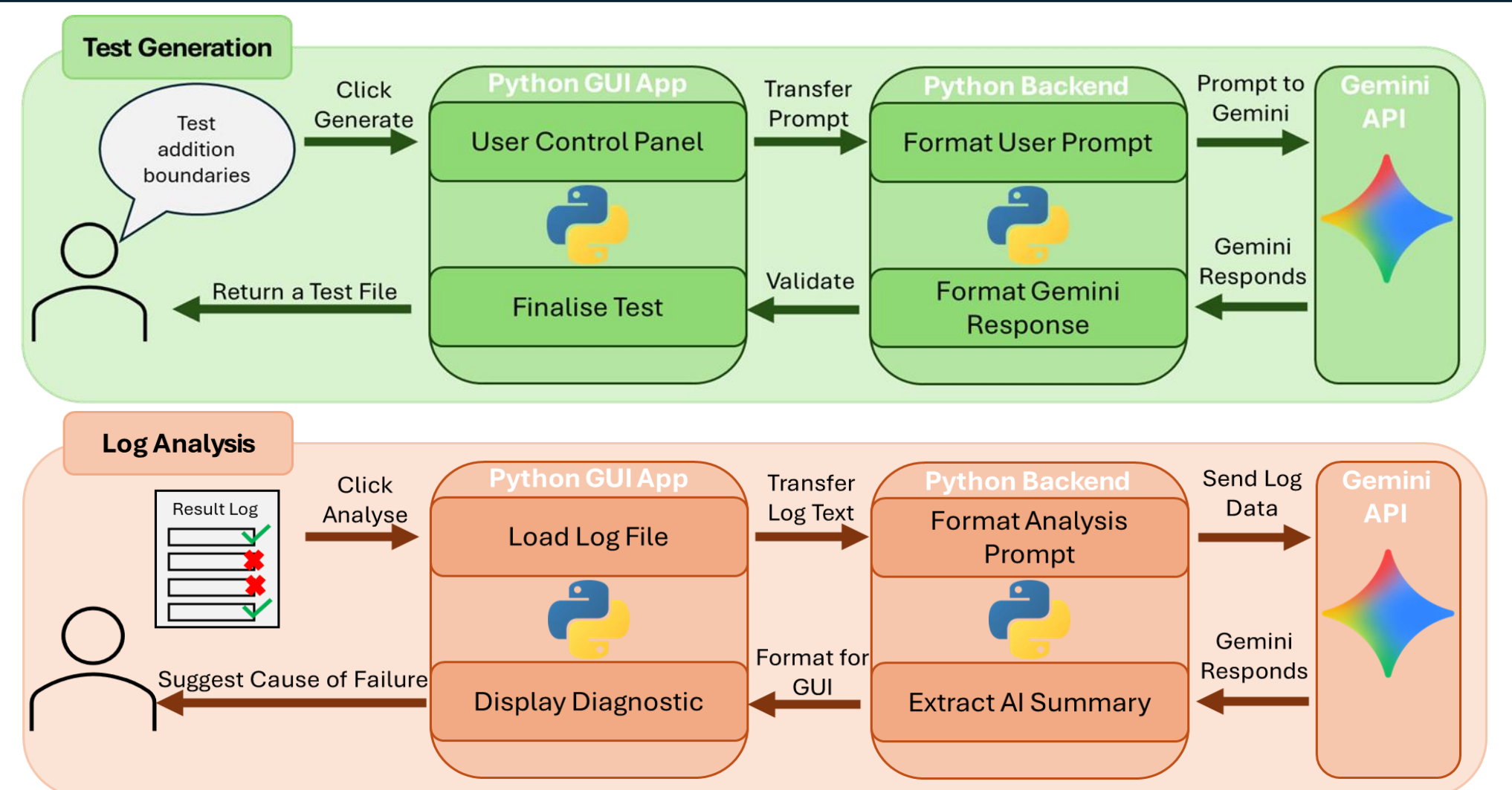
| XE636 Individual Project Gantt Chart                       |            |            |                 |  | 13 Oct 2025 | 20 Oct 2025 | 27 Oct 2025 | 3 Nov 2025 | 10 Nov 2025 | 17 Nov 2025 | 24 Nov 2025 | 1 Dec 2025 | 8 Dec 2025 | 15 Dec 2025 | 22 Dec 2025 | 29 Dec 2025 | 5 Jan 2026 | 12 Jan 2026 | 19 Jan 2026 | 26 Jan 2026 | 2 Feb 2026 | 9 Feb 2026 | 16 Feb 2026 | 23 Feb 2026 | 2 Mar 2026 | 9 Mar 2026 | 16 Mar 2026 | 23 Mar 2026 | 30 Mar 2026 | 6 Apr 2026 | 13 Apr 2026 | 20 Apr 2026 | 27 Apr 2026 | 4 May 2026 | 11 May 2026 |  |  |  |
|------------------------------------------------------------|------------|------------|-----------------|--|-------------|-------------|-------------|------------|-------------|-------------|-------------|------------|------------|-------------|-------------|-------------|------------|-------------|-------------|-------------|------------|------------|-------------|-------------|------------|------------|-------------|-------------|-------------|------------|-------------|-------------|-------------|------------|-------------|--|--|--|
|                                                            |            |            | Mon, 06/10/2025 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
|                                                            |            |            | 1               |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| TASK                                                       | START      | END        |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| Phase 1: Formalisation & Planning                          | 13/10/2025 | 19/11/2025 |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| Milestone M1: A01 Scoping Report Submitted                 | 19/11/25   | 19/11/25   |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| Phase 2: Advanced Development & Formative Review           | 20/11/25   | 16/1/26    |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| Milestone M2: A02 Formative Report Submitted               | 12/12/25   | 12/12/25   |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| Milestone M3: A02 Formative Viva Completed                 | 16/1/26    | 16/1/26    |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| Phase 3: Full System Integration, Benchmarking & Poster    | 19/1/26    | 24/4/26    |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| Milestone M4: Full H/W S/W System Integrated & Verified    | 20/2/26    | 20/2/26    |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| Milestone M5: A03 Poster Submitted for University Printing | 20/4/26    | 20/4/26    |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| Milestone M6: A03 Poster Submitted to Turnitin             | 22/4/26    | 22/4/26    |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| Milestone M7: A03 Poster Exhibition Day                    | 24/4/26    | 24/4/26    |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| Phase 4: Dissertation Write-Up & Final Viva                | 1/4/26     | 22/5/26    |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| Milestone M8: A04 Final Dissertation Submitted             | 1/5/26     | 1/5/26     |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |
| Milestone M9: A04 Final Viva-voce Completed                | 22/5/26    | 22/5/26    |                 |  |             |             |             |            |             |             |             |            |            |             |             |             |            |             |             |             |            |            |             |             |            |            |             |             |             |            |             |             |             |            |             |  |  |  |

## 3 System Architecture

- PC Controller:** Python middleware automatically parses inputs and logs results.
- Interface:** Connects systems via a physical USB to UART adapter.
- FPGA Testbed:** Decodes commands and tests the 16-bit ALU.



## 5 Artificial Intelligence Integration



## 7 Wider Context

- Environment:** Mitigates e-waste by using standard boards. The ecological footprint of cloud AI is offset through selective querying, ensuring the resources saved by preventing physical bugs outweigh data centre costs.
- Ethics:** Aligning with the IET Rules of Conduct [4], AI is deployed strictly as an augmentation tool. It mandates rigorous human oversight to ensure efficiency gains never compromise safety or accountability.
- Society:** Automation drastically reduces engineering fatigue and accelerates commercial time-to-market but requires balanced implementation to address valid concerns regarding junior workforce displacement.
- Legal & IP:** The framework is developed as open-source. However, relying on public cloud APIs requires transmitting behavioural hardware logs off-site, introducing IP security risks. Future enterprise iterations must integrate local, offline LLMs to ensure strict commercial data privacy.

[1] Siemens EDA, "2022 Wilson Research Group FPGA functional verification trends." 2022.

[2] F. Mihaljević, M. Trunčević, and A. Iren, "Hardware-in-the-Loop Simulations: A Historical Overview of Engineering Challenges," MDPI, Aug. 2022.

[3] S. Jiang, C. Torng, and C. Batten, "An Open-Source Python-Based Hardware Generation, Simulation, and Verification Framework," 2018.

[4] The Institution of Engineering and Technology, "Rules of Conduct," [Online]. Available: <https://www.theiet.org/about/governance/rules-of-conduct>.